

# Complete Results of *Parallelizing Node-Level Explainability in Graph Neural Networks*

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## 1. Complete results

This document presents the complete set of results for the paper *Parallelizing Node-Level Explainability in Graph Neural Networks*. These results were obtained for two models (GCN [1] and GAT [2]) using four explainability techniques (DeconvNet [3], GNNExplainer [4], Guided Backprop [5], and Saliency Maps [6]) on three citation network datasets: Cora, CiteSeer, and PubMed [7].

Table 1 serves as an index to all detailed result tables corresponding to each combination of explainability technique, dataset, and model architecture. For every combination, we evaluate multiple clustering configurations (ranging from 1 to 128 clusters) and vary the dropout rate between 0 and 1. Reported metrics include execution time, increases in nodes and edges, and the percentage of affected explanations and nodes.

Note that for the PubMed dataset, which is significantly larger than Cora and CiteSeer, experiments are limited to 16 clusters due to the GPU memory required to store the reconstructed dataset. Running larger cluster configurations on CPU would prevent a fair comparison with the GPU-based experiments on the smaller datasets, since CPU execution times would be

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disproportionately higher. Similarly, GNNExplainer is not run on PubMed because its memory requirements exceed the available GPU capacity.

Table 1: Summary of table references for each explainability technique evaluated on the Cora, CiteSeer, and PubMed datasets under GCN and GAT model architectures.

| Explainability  | Cora    |         | CiteSeer |          | PubMed   |          |
|-----------------|---------|---------|----------|----------|----------|----------|
| technique       | GCN     | GAT     | GCN      | GAT      | GCN      | GAT      |
| DeconvNet       | Table 2 | Table 6 | Table 10 | Table 14 | Table 18 | Table 21 |
| GNNExplainer    | Table 3 | Table 7 | Table 11 | Table 15 | –        | –        |
| Guided Backprop | Table 4 | Table 8 | Table 12 | Table 16 | Table 19 | Table 22 |
| Saliency Map    | Table 5 | Table 9 | Table 13 | Table 17 | Table 20 | Table 23 |

## References

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Table 2: Dropout reconstruction with DeconvNet on Cora dataset using the GCN model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 17.01                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 3.07                  | 720.35                | 774.97                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.61                  | 576.00                | 498.22                | 10.78                        | 1.14  | 0.37  | 0.46                  | 0.03 | 0.01 |
|                       | 0.50            | 1.94                  | 361.63                | 182.91                | 22.01                        | 2.73  | 0.41  | 1.20                  | 0.08 | 0.01 |
|                       | 0.70            | 1.58                  | 218.28                | 60.21                 | 27.66                        | 4.21  | 1.33  | 1.64                  | 0.12 | 0.04 |
|                       | 1.00            | 1.34                  | 0.00                  | -13.93                | 31.09                        | 5.83  | 1.44  | 2.04                  | 0.17 | 0.04 |
| 32                    | 0.00            | 2.10                  | 1084.86               | 1149.77               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.75                  | 871.05                | 729.23                | 14.99                        | 2.10  | 0.48  | 0.70                  | 0.06 | 0.01 |
|                       | 0.50            | 1.29                  | 544.09                | 291.68                | 26.66                        | 4.14  | 0.96  | 1.52                  | 0.12 | 0.03 |
|                       | 0.70            | 1.00                  | 323.38                | 96.23                 | 32.61                        | 5.32  | 1.18  | 2.09                  | 0.16 | 0.03 |
|                       | 1.00            | 0.84                  | 0.00                  | -18.57                | 37.92                        | 7.05  | 1.92  | 2.63                  | 0.21 | 0.05 |
| 64                    | 0.00            | 1.61                  | 1667.25               | 1736.23               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.41                  | 1335.19               | 1109.95               | 17.36                        | 1.70  | 0.26  | 0.77                  | 0.05 | 0.01 |
|                       | 0.50            | 1.04                  | 838.07                | 424.99                | 33.83                        | 4.25  | 0.89  | 2.07                  | 0.13 | 0.02 |
|                       | 0.70            | 0.89                  | 498.71                | 139.39                | 39.81                        | 6.39  | 1.44  | 2.77                  | 0.19 | 0.04 |
|                       | 1.00            | 0.65                  | 0.00                  | -24.40                | 45.46                        | 8.42  | 2.10  | 3.57                  | 0.26 | 0.06 |
| 128                   | 0.00            | 1.64                  | 2774.52               | 2860.34               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.44                  | 2221.42               | 1827.06               | 34.42                        | 9.49  | 2.81  | 1.83                  | 0.28 | 0.08 |
|                       | 0.50            | 1.16                  | 1385.89               | 710.21                | 56.91                        | 20.57 | 8.38  | 4.80                  | 0.69 | 0.24 |
|                       | 0.70            | 0.95                  | 829.28                | 236.55                | 64.07                        | 27.10 | 10.86 | 6.60                  | 0.96 | 0.32 |
|                       | 1.00            | 0.78                  | 0.00                  | -50.42                | 72.12                        | 36.48 | 14.92 | 8.65                  | 1.43 | 0.47 |

Table 3: Dropout reconstruction with GNNExplainer on the Cora dataset using the GCN model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations ( %) |       |       | Affected<br>nodes (%) |       |       |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|-------------------------------|-------|-------|-----------------------|-------|-------|
|                       |                 |                       |                       |                       | 0.2                           | 0.5   | 0.7   | 0.2                   | 0.5   | 0.7   |
| 1                     | 0.00            | 1903.56               | 0.00                  | 0.00                  | 0.00                          | 0.00  | 0.00  | 0.00                  | 0.00  | 0.00  |
| 16                    | 0.00            | 299.03                | 720.35                | 774.97                | 83.12                         | 49.37 | 0.00  | 12.22                 | 4.60  | 0.00  |
|                       | 0.20            | 242.98                | 579.39                | 504.32                | 91.03                         | 57.75 | 19.09 | 22.33                 | 6.07  | 1.30  |
|                       | 0.50            | 170.69                | 359.38                | 188.73                | 92.32                         | 60.89 | 31.31 | 39.49                 | 8.74  | 3.55  |
|                       | 0.70            | 141.48                | 214.40                | 66.62                 | 92.54                         | 61.34 | 34.64 | 47.10                 | 9.66  | 4.42  |
|                       | 1.00            | 120.62                | 0.00                  | -13.93                | 92.36                         | 60.86 | 38.74 | 54.10                 | 10.80 | 5.44  |
| 32                    | 0.00            | 194.01                | 1084.86               | 1149.77               | 84.34                         | 51.29 | 0.00  | 12.56                 | 4.76  | 0.00  |
|                       | 0.20            | 154.45                | 869.35                | 741.93                | 92.02                         | 61.08 | 26.81 | 26.18                 | 7.12  | 2.25  |
|                       | 0.50            | 103.15                | 538.26                | 281.17                | 92.13                         | 62.44 | 37.56 | 46.51                 | 9.97  | 5.05  |
|                       | 0.70            | 76.90                 | 328.32                | 95.55                 | 92.21                         | 61.23 | 41.77 | 56.31                 | 11.45 | 6.39  |
|                       | 1.00            | 59.86                 | 0.00                  | -18.57                | 92.69                         | 62.41 | 45.79 | 66.08                 | 13.06 | 7.89  |
| 64                    | 0.00            | 130.85                | 1667.25               | 1736.23               | 85.78                         | 53.84 | 0.00  | 12.96                 | 5.06  | 0.00  |
|                       | 0.20            | 101.38                | 1331.68               | 1113.70               | 93.46                         | 62.48 | 31.24 | 32.56                 | 8.15  | 2.82  |
|                       | 0.50            | 63.74                 | 831.06                | 428.15                | 93.98                         | 64.81 | 43.54 | 52.59                 | 11.30 | 6.03  |
|                       | 0.70            | 45.04                 | 505.39                | 152.39                | 92.87                         | 64.25 | 47.19 | 61.66                 | 12.93 | 7.83  |
|                       | 1.00            | 29.70                 | 0.00                  | -24.40                | 92.50                         | 64.33 | 51.85 | 74.85                 | 15.02 | 9.92  |
| 128                   | 0.00            | 100.70                | 2774.52               | 2860.34               | 85.71                         | 54.76 | 0.00  | 12.91                 | 5.05  | 0.00  |
|                       | 0.20            | 75.67                 | 2221.57               | 1826.53               | 94.98                         | 68.21 | 42.91 | 37.19                 | 9.34  | 4.16  |
|                       | 0.50            | 45.38                 | 1384.93               | 709.26                | 94.94                         | 71.27 | 57.87 | 62.70                 | 14.22 | 9.51  |
|                       | 0.70            | 29.81                 | 838.33                | 251.06                | 94.53                         | 73.08 | 63.85 | 73.83                 | 16.77 | 12.37 |
|                       | 1.00            | 15.51                 | 0.00                  | -50.42                | 92.39                         | 76.55 | 68.80 | 88.14                 | 20.11 | 15.82 |

Table 4: Dropout reconstruction with Guided Backprop on the Cora dataset using the GCN model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 16.82                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 3.09                  | 720.35                | 774.97                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.61                  | 573.38                | 487.86                | 12.37                        | 0.96  | 0.30  | 0.54                  | 0.03 | 0.01 |
|                       | 0.50            | 1.93                  | 361.26                | 191.27                | 21.57                        | 2.95  | 0.63  | 1.17                  | 0.08 | 0.02 |
|                       | 0.70            | 1.59                  | 215.58                | 59.97                 | 27.62                        | 3.99  | 0.96  | 1.65                  | 0.11 | 0.03 |
|                       | 1.00            | 1.31                  | 0.00                  | -13.93                | 31.06                        | 5.83  | 1.44  | 2.04                  | 0.17 | 0.04 |
| 32                    | 0.00            | 2.13                  | 1084.86               | 1149.77               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.76                  | 869.57                | 727.89                | 14.07                        | 1.66  | 0.33  | 0.66                  | 0.05 | 0.01 |
|                       | 0.50            | 1.29                  | 549.30                | 280.58                | 26.55                        | 4.25  | 1.11  | 1.58                  | 0.12 | 0.03 |
|                       | 0.70            | 1.02                  | 324.89                | 101.25                | 33.35                        | 5.24  | 1.33  | 2.13                  | 0.15 | 0.04 |
|                       | 1.00            | 0.80                  | 0.00                  | -18.57                | 37.89                        | 7.05  | 1.92  | 2.64                  | 0.21 | 0.05 |
| 64                    | 0.00            | 1.64                  | 1667.25               | 1736.23               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.41                  | 1331.54               | 1105.49               | 18.72                        | 1.92  | 0.22  | 0.88                  | 0.05 | 0.01 |
|                       | 0.50            | 1.03                  | 833.09                | 422.87                | 32.83                        | 4.39  | 0.89  | 2.02                  | 0.13 | 0.02 |
|                       | 0.70            | 0.85                  | 498.63                | 147.82                | 38.66                        | 6.09  | 1.55  | 2.70                  | 0.19 | 0.04 |
|                       | 1.00            | 0.69                  | 0.00                  | -24.40                | 45.49                        | 8.42  | 2.10  | 3.57                  | 0.26 | 0.06 |
| 128                   | 0.00            | 1.57                  | 2774.52               | 2860.34               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.44                  | 2221.09               | 1844.18               | 34.97                        | 8.42  | 3.03  | 1.85                  | 0.25 | 0.09 |
|                       | 0.50            | 1.11                  | 1390.18               | 705.12                | 57.02                        | 20.49 | 7.75  | 4.73                  | 0.67 | 0.22 |
|                       | 0.70            | 0.99                  | 828.29                | 230.49                | 63.63                        | 25.78 | 10.27 | 6.57                  | 0.92 | 0.31 |
|                       | 1.00            | 0.78                  | 0.00                  | -50.42                | 72.08                        | 36.48 | 14.92 | 8.65                  | 1.43 | 0.47 |

Table 5: Dropout reconstruction with Saliency Map on Cora dataset using the GCN model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 16.96                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 2.89                  | 720.35                | 774.97                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.49                  | 576.00                | 498.22                | 10.78                        | 1.14  | 0.41  | 0.46                  | 0.03 | 0.01 |
|                       | 0.50            | 1.85                  | 361.63                | 182.91                | 22.23                        | 3.06  | 0.44  | 1.22                  | 0.09 | 0.01 |
|                       | 0.70            | 1.56                  | 218.28                | 60.21                 | 27.92                        | 4.36  | 1.40  | 1.66                  | 0.13 | 0.04 |
|                       | 1.00            | 1.33                  | 0.00                  | -13.93                | 31.28                        | 6.13  | 1.51  | 2.05                  | 0.18 | 0.04 |
| 32                    | 0.00            | 2.02                  | 1084.86               | 1149.77               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.66                  | 871.05                | 729.23                | 15.14                        | 2.18  | 0.52  | 0.71                  | 0.06 | 0.01 |
|                       | 0.50            | 1.23                  | 544.09                | 291.68                | 26.62                        | 4.43  | 1.03  | 1.52                  | 0.13 | 0.03 |
|                       | 0.70            | 0.99                  | 323.38                | 96.23                 | 32.75                        | 5.58  | 1.26  | 2.09                  | 0.16 | 0.04 |
|                       | 1.00            | 0.81                  | 0.00                  | -18.57                | 37.92                        | 7.42  | 2.03  | 2.62                  | 0.22 | 0.06 |
| 64                    | 0.00            | 1.55                  | 1667.25               | 1736.23               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.33                  | 1335.19               | 1109.95               | 17.32                        | 1.81  | 0.26  | 0.76                  | 0.05 | 0.01 |
|                       | 0.50            | 1.00                  | 838.07                | 424.99                | 33.97                        | 4.51  | 0.92  | 2.06                  | 0.13 | 0.03 |
|                       | 0.70            | 0.83                  | 498.71                | 139.39                | 39.88                        | 6.61  | 1.48  | 2.78                  | 0.20 | 0.04 |
|                       | 1.00            | 0.71                  | 0.00                  | -24.40                | 45.64                        | 8.79  | 2.18  | 3.58                  | 0.27 | 0.06 |
| 128                   | 0.00            | 1.52                  | 2774.52               | 2860.34               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.37                  | 2221.42               | 1827.06               | 34.42                        | 9.60  | 2.88  | 1.82                  | 0.29 | 0.08 |
|                       | 0.50            | 1.08                  | 1385.89               | 710.21                | 56.65                        | 20.75 | 8.60  | 4.74                  | 0.70 | 0.25 |
|                       | 0.70            | 0.98                  | 829.28                | 236.55                | 63.81                        | 27.33 | 11.08 | 6.56                  | 0.97 | 0.33 |
|                       | 1.00            | 0.79                  | 0.00                  | -50.42                | 72.16                        | 36.74 | 15.32 | 8.58                  | 1.44 | 0.48 |

Table 6: Dropout reconstruction with DeconvNet on the Cora dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 27.32                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 4.47                  | 720.35                | 774.97                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 3.80                  | 579.06                | 512.77                | 9.45                         | 6.79  | 6.09  | 0.82                  | 0.40 | 0.29 |
|                       | 0.50            | 2.79                  | 364.62                | 191.82                | 23.23                        | 16.65 | 14.55 | 2.12                  | 0.96 | 0.67 |
|                       | 0.70            | 2.44                  | 215.03                | 65.95                 | 30.50                        | 22.90 | 20.46 | 2.86                  | 1.40 | 1.02 |
|                       | 1.00            | 2.06                  | 0.00                  | -13.93                | 36.19                        | 27.03 | 24.34 | 3.47                  | 1.65 | 1.22 |
| 32                    | 0.00            | 2.87                  | 1084.86               | 1149.77               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.41                  | 866.25                | 745.26                | 13.29                        | 7.20  | 5.83  | 0.90                  | 0.37 | 0.25 |
|                       | 0.50            | 1.76                  | 541.54                | 285.13                | 30.80                        | 23.19 | 20.49 | 2.92                  | 1.41 | 1.03 |
|                       | 0.70            | 1.48                  | 329.87                | 99.79                 | 39.84                        | 29.95 | 27.25 | 3.86                  | 1.87 | 1.36 |
|                       | 1.00            | 1.17                  | 0.00                  | -18.57                | 45.42                        | 34.49 | 31.65 | 4.52                  | 2.22 | 1.62 |
| 64                    | 0.00            | 2.14                  | 1667.25               | 1736.23               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.83                  | 1334.01               | 1117.39               | 19.17                        | 13.52 | 11.89 | 1.60                  | 0.78 | 0.55 |
|                       | 0.50            | 1.38                  | 831.54                | 418.85                | 38.11                        | 28.10 | 25.00 | 3.40                  | 1.61 | 1.18 |
|                       | 0.70            | 1.07                  | 498.67                | 141.72                | 44.72                        | 33.97 | 30.35 | 4.25                  | 2.00 | 1.45 |
|                       | 1.00            | 0.85                  | 0.00                  | -24.40                | 54.43                        | 43.02 | 39.07 | 5.50                  | 2.70 | 1.97 |
| 128                   | 0.00            | 1.98                  | 2774.52               | 2860.34               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.80                  | 2226.29               | 1830.98               | 29.58                        | 20.16 | 17.17 | 2.37                  | 1.11 | 0.79 |
|                       | 0.50            | 1.31                  | 1388.70               | 705.51                | 50.30                        | 37.44 | 32.83 | 4.71                  | 2.26 | 1.61 |
|                       | 0.70            | 1.10                  | 842.87                | 233.93                | 61.63                        | 51.22 | 47.23 | 6.48                  | 3.28 | 2.43 |
|                       | 1.00            | 0.88                  | 0.00                  | -50.42                | 70.35                        | 60.71 | 56.20 | 7.38                  | 3.94 | 2.96 |

Table 7: Dropout reconstruction with GNNExplainer on the Cora dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |       |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|-------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5   | 0.7  |
| 1                     | 0.00            | 3086.39               | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00  | 0.00 |
| 16                    | 0.00            | 421.13                | 720.35                | 774.97                | 73.86                        | 31.17 | 0.11  | 7.37                  | 1.40  | 0.00 |
|                       | 0.20            | 344.39                | 575.74                | 495.36                | 87.63                        | 43.06 | 18.32 | 18.80                 | 2.73  | 0.93 |
|                       | 0.50            | 256.32                | 362.48                | 188.82                | 89.07                        | 49.56 | 28.40 | 31.97                 | 4.26  | 2.00 |
|                       | 0.70            | 219.19                | 214.14                | 62.16                 | 89.62                        | 54.17 | 36.48 | 43.11                 | 5.64  | 2.99 |
|                       | 1.00            | 193.44                | 0.00                  | -13.93                | 90.58                        | 56.65 | 40.99 | 50.99                 | 6.58  | 3.63 |
| 32                    | 0.00            | 269.67                | 1084.86               | 1149.77               | 74.52                        | 31.39 | 0.04  | 7.33                  | 1.42  | 0.00 |
|                       | 0.20            | 216.96                | 871.86                | 745.40                | 86.48                        | 45.13 | 22.01 | 25.12                 | 3.28  | 1.39 |
|                       | 0.50            | 149.55                | 544.87                | 287.32                | 90.47                        | 54.65 | 37.48 | 41.30                 | 5.28  | 2.80 |
|                       | 0.70            | 118.61                | 324.08                | 89.52                 | 91.06                        | 58.42 | 44.94 | 54.91                 | 6.90  | 4.04 |
|                       | 1.00            | 98.50                 | 0.00                  | -18.57                | 91.62                        | 61.30 | 49.82 | 63.73                 | 7.93  | 4.85 |
| 64                    | 0.00            | 180.51                | 1667.25               | 1736.23               | 76.22                        | 32.27 | 0.07  | 7.50                  | 1.48  | 0.00 |
|                       | 0.20            | 140.31                | 1333.09               | 1128.76               | 91.14                        | 51.51 | 28.80 | 23.66                 | 3.58  | 1.47 |
|                       | 0.50            | 91.81                 | 835.71                | 425.50                | 92.43                        | 60.30 | 46.27 | 51.35                 | 6.67  | 3.94 |
|                       | 0.70            | 67.10                 | 504.03                | 149.37                | 92.17                        | 61.67 | 49.96 | 61.05                 | 7.71  | 4.67 |
|                       | 1.00            | 47.99                 | 0.00                  | -24.40                | 91.17                        | 65.77 | 57.05 | 73.13                 | 9.04  | 5.92 |
| 128                   | 0.00            | 140.68                | 2774.52               | 2860.34               | 76.88                        | 32.64 | 0.00  | 7.60                  | 1.47  | 0.00 |
|                       | 0.20            | 105.35                | 2220.27               | 1823.51               | 93.80                        | 59.68 | 39.73 | 34.10                 | 5.01  | 2.54 |
|                       | 0.50            | 63.99                 | 1385.19               | 695.95                | 93.94                        | 68.80 | 59.23 | 63.42                 | 8.79  | 5.62 |
|                       | 0.70            | 42.99                 | 832.72                | 230.73                | 92.95                        | 72.45 | 65.66 | 77.03                 | 10.61 | 7.25 |
|                       | 1.00            | 24.76                 | 0.00                  | -50.42                | 92.13                        | 75.96 | 71.05 | 87.54                 | 12.59 | 8.97 |

Table 8: Dropout reconstruction with Guided Backprop on the Cora dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 27.72                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 4.38                  | 720.35                | 774.97                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 3.69                  | 577.36                | 492.12                | 9.05                         | 5.98  | 5.10  | 0.76                  | 0.34 | 0.23 |
|                       | 0.50            | 2.79                  | 359.75                | 187.86                | 23.49                        | 16.43 | 14.77 | 2.07                  | 0.97 | 0.70 |
|                       | 0.70            | 2.41                  | 218.02                | 58.83                 | 31.50                        | 23.49 | 21.86 | 3.02                  | 1.49 | 1.11 |
|                       | 1.00            | 2.12                  | 0.00                  | -13.93                | 35.82                        | 26.99 | 24.37 | 3.44                  | 1.65 | 1.23 |
| 32                    | 0.00            | 2.91                  | 1084.86               | 1149.77               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.44                  | 868.43                | 749.43                | 15.07                        | 9.64  | 8.35  | 1.19                  | 0.52 | 0.38 |
|                       | 0.50            | 1.78                  | 547.27                | 286.36                | 27.77                        | 20.49 | 18.21 | 2.67                  | 1.25 | 0.89 |
|                       | 0.70            | 1.49                  | 324.48                | 94.30                 | 34.27                        | 25.00 | 21.79 | 3.17                  | 1.51 | 1.07 |
|                       | 1.00            | 1.20                  | 0.00                  | -18.57                | 45.05                        | 34.75 | 31.68 | 4.49                  | 2.22 | 1.62 |
| 64                    | 0.00            | 2.14                  | 1667.25               | 1736.23               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.85                  | 1331.24               | 1088.97               | 18.57                        | 12.59 | 11.37 | 1.60                  | 0.74 | 0.55 |
|                       | 0.50            | 1.33                  | 835.27                | 436.40                | 33.09                        | 23.23 | 20.53 | 3.02                  | 1.36 | 1.01 |
|                       | 0.70            | 1.12                  | 501.85                | 147.59                | 40.32                        | 29.73 | 26.66 | 3.84                  | 1.78 | 1.29 |
|                       | 1.00            | 0.89                  | 0.00                  | -24.40                | 54.17                        | 42.98 | 38.88 | 5.46                  | 2.69 | 1.97 |
| 128                   | 0.00            | 1.99                  | 2774.52               | 2860.34               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.78                  | 2216.32               | 1840.83               | 23.67                        | 15.44 | 12.70 | 1.93                  | 0.85 | 0.59 |
|                       | 0.50            | 1.37                  | 1395.27               | 722.85                | 49.89                        | 38.44 | 34.71 | 4.93                  | 2.37 | 1.70 |
|                       | 0.70            | 1.10                  | 833.09                | 226.75                | 59.60                        | 48.74 | 44.61 | 6.26                  | 3.12 | 2.31 |
|                       | 1.00            | 0.89                  | 0.00                  | -50.42                | 70.24                        | 60.64 | 56.31 | 7.36                  | 3.92 | 2.95 |

Table 9: Dropout reconstruction with Saliency Map on the Cora dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 28.40                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 4.30                  | 720.35                | 774.97                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 3.62                  | 579.25                | 506.63                | 10.64                        | 5.91  | 4.69  | 0.82                  | 0.31 | 0.21 |
|                       | 0.50            | 2.83                  | 360.78                | 188.27                | 20.57                        | 13.59 | 11.56 | 1.85                  | 0.77 | 0.50 |
|                       | 0.70            | 2.39                  | 215.44                | 57.09                 | 32.83                        | 24.67 | 21.75 | 3.15                  | 1.42 | 0.95 |
|                       | 1.00            | 2.06                  | 0.00                  | -13.93                | 37.22                        | 27.70 | 24.78 | 3.59                  | 1.62 | 1.09 |
| 32                    | 0.00            | 2.91                  | 1084.86               | 1149.77               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.45                  | 866.58                | 741.32                | 15.55                        | 9.71  | 8.09  | 1.32                  | 0.55 | 0.36 |
|                       | 0.50            | 1.78                  | 542.73                | 276.85                | 27.95                        | 18.13 | 15.66 | 2.38                  | 0.98 | 0.68 |
|                       | 0.70            | 1.43                  | 321.45                | 86.21                 | 37.00                        | 26.00 | 23.01 | 3.53                  | 1.48 | 1.04 |
|                       | 1.00            | 1.18                  | 0.00                  | -18.57                | 46.09                        | 34.86 | 31.28 | 4.57                  | 2.10 | 1.46 |
| 64                    | 0.00            | 2.18                  | 1667.25               | 1736.23               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.83                  | 1333.68               | 1110.21               | 18.54                        | 12.63 | 10.89 | 1.50                  | 0.68 | 0.46 |
|                       | 0.50            | 1.33                  | 831.65                | 416.88                | 35.52                        | 24.74 | 21.01 | 3.23                  | 1.37 | 0.94 |
|                       | 0.70            | 1.07                  | 503.47                | 146.82                | 45.86                        | 34.42 | 30.61 | 4.43                  | 1.97 | 1.38 |
|                       | 1.00            | 0.84                  | 0.00                  | -24.40                | 54.43                        | 43.24 | 38.55 | 5.60                  | 2.59 | 1.79 |
| 128                   | 0.00            | 2.05                  | 2774.52               | 2860.34               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.77                  | 2217.87               | 1829.73               | 28.99                        | 19.09 | 15.69 | 2.31                  | 1.00 | 0.67 |
|                       | 0.50            | 1.31                  | 1387.04               | 701.04                | 52.62                        | 39.00 | 34.01 | 5.11                  | 2.28 | 1.55 |
|                       | 0.70            | 1.12                  | 831.35                | 230.73                | 63.81                        | 50.41 | 45.57 | 6.47                  | 3.10 | 2.16 |
|                       | 1.00            | 0.88                  | 0.00                  | -50.42                | 71.05                        | 60.45 | 55.61 | 7.59                  | 3.87 | 2.79 |

Table 10: Dropout reconstruction with DeconvNet on the CiteSeer dataset using the GCN model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |      | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7  | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 21.56                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 2.40                  | 144.27                | 216.10                | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.24                  | 115.33                | 133.48                | 3.88                         | 0.51  | 0.21 | 0.42                  | 0.03 | 0.01 |
|                       | 0.50            | 1.93                  | 72.86                 | 54.57                 | 7.69                         | 1.20  | 0.36 | 0.97                  | 0.08 | 0.02 |
|                       | 0.70            | 1.80                  | 43.70                 | 15.55                 | 10.31                        | 1.50  | 0.51 | 1.39                  | 0.11 | 0.03 |
|                       | 1.00            | 1.67                  | 0.00                  | -6.72                 | 11.72                        | 2.31  | 0.81 | 1.68                  | 0.17 | 0.05 |
| 32                    | 0.00            | 1.57                  | 244.27                | 372.58                | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.44                  | 194.98                | 240.95                | 4.87                         | 0.48  | 0.15 | 0.53                  | 0.03 | 0.01 |
|                       | 0.50            | 1.25                  | 119.81                | 89.81                 | 10.46                        | 1.26  | 0.24 | 1.40                  | 0.09 | 0.02 |
|                       | 0.70            | 1.09                  | 72.11                 | 27.59                 | 13.04                        | 2.01  | 0.39 | 1.93                  | 0.14 | 0.03 |
|                       | 1.00            | 0.96                  | 0.00                  | -9.47                 | 15.18                        | 2.58  | 0.57 | 2.27                  | 0.18 | 0.04 |
| 64                    | 0.00            | 1.21                  | 383.68                | 582.29                | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.12                  | 306.94                | 364.48                | 7.87                         | 1.17  | 0.39 | 0.89                  | 0.09 | 0.03 |
|                       | 0.50            | 0.95                  | 195.94                | 142.82                | 13.20                        | 1.92  | 0.54 | 1.86                  | 0.14 | 0.04 |
|                       | 0.70            | 0.88                  | 113.13                | 45.47                 | 15.72                        | 2.37  | 0.75 | 2.49                  | 0.18 | 0.05 |
|                       | 1.00            | 0.73                  | 0.00                  | -13.16                | 18.52                        | 3.40  | 1.05 | 3.23                  | 0.25 | 0.07 |
| 128                   | 0.00            | 1.23                  | 647.40                | 1012.32               | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.32                  | 520.47                | 649.30                | 14.10                        | 3.07  | 0.90 | 1.70                  | 0.21 | 0.06 |
|                       | 0.50            | 1.10                  | 324.29                | 251.43                | 24.17                        | 7.75  | 2.71 | 4.55                  | 0.60 | 0.19 |
|                       | 0.70            | 0.99                  | 192.61                | 73.95                 | 28.16                        | 10.37 | 3.52 | 6.45                  | 0.89 | 0.25 |
|                       | 1.00            | 0.90                  | 0.00                  | -29.09                | 32.37                        | 14.43 | 5.50 | 8.40                  | 1.34 | 0.40 |

Table 11: Dropout reconstruction with GNNExplainer on the CiteSeer dataset using the GCN model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |       |       |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|-------|-------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5   | 0.7   |
| 1                     | 0.00            | 2340.40               | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00  | 0.00  |
| 16                    | 0.00            | 211.36                | 144.27                | 216.10                | 72.50                        | 41.03 | 0.00  | 18.96                 | 9.07  | 0.00  |
|                       | 0.20            | 190.66                | 114.55                | 138.44                | 74.66                        | 42.20 | 7.27  | 25.03                 | 9.95  | 1.09  |
|                       | 0.50            | 169.88                | 72.20                 | 51.80                 | 75.08                        | 43.13 | 10.34 | 32.68                 | 11.15 | 2.22  |
|                       | 0.70            | 156.77                | 42.35                 | 13.42                 | 74.96                        | 42.17 | 13.04 | 38.60                 | 11.62 | 3.12  |
|                       | 1.00            | 148.35                | 0.00                  | -6.72                 | 74.78                        | 41.87 | 14.76 | 41.97                 | 12.08 | 3.80  |
| 32                    | 0.00            | 133.41                | 244.27                | 372.58                | 73.34                        | 42.35 | 0.00  | 19.09                 | 9.30  | 0.00  |
|                       | 0.20            | 115.96                | 196.54                | 235.76                | 75.68                        | 42.89 | 8.78  | 28.17                 | 10.42 | 1.45  |
|                       | 0.50            | 93.50                 | 124.89                | 91.41                 | 75.68                        | 42.05 | 13.74 | 39.82                 | 11.96 | 3.47  |
|                       | 0.70            | 83.97                 | 74.33                 | 27.24                 | 75.56                        | 41.90 | 16.44 | 44.57                 | 12.64 | 4.36  |
|                       | 1.00            | 74.45                 | 0.00                  | -9.47                 | 74.51                        | 42.02 | 18.46 | 50.18                 | 13.44 | 5.42  |
| 64                    | 0.00            | 83.20                 | 383.68                | 582.29                | 73.25                        | 42.47 | 0.00  | 19.02                 | 9.45  | 0.00  |
|                       | 0.20            | 69.99                 | 304.75                | 357.82                | 76.16                        | 42.89 | 10.40 | 29.09                 | 10.68 | 1.98  |
|                       | 0.50            | 52.62                 | 193.57                | 144.31                | 74.60                        | 41.99 | 15.75 | 43.06                 | 12.41 | 4.50  |
|                       | 0.70            | 44.66                 | 115.66                | 43.65                 | 74.27                        | 40.91 | 17.88 | 50.67                 | 13.52 | 6.08  |
|                       | 1.00            | 37.24                 | 0.00                  | -13.16                | 73.34                        | 40.64 | 20.02 | 57.08                 | 14.54 | 7.63  |
| 128                   | 0.00            | 58.97                 | 647.40                | 1012.32               | 73.61                        | 41.99 | 0.00  | 19.09                 | 9.47  | 0.00  |
|                       | 0.20            | 48.52                 | 519.21                | 653.16                | 78.06                        | 44.24 | 15.81 | 34.32                 | 11.49 | 3.07  |
|                       | 0.50            | 34.22                 | 323.23                | 241.34                | 75.98                        | 44.42 | 24.05 | 52.47                 | 14.99 | 7.94  |
|                       | 0.70            | 26.19                 | 198.14                | 79.57                 | 74.48                        | 43.76 | 27.47 | 61.00                 | 16.39 | 10.28 |
|                       | 1.00            | 18.92                 | 0.00                  | -29.09                | 72.38                        | 44.12 | 30.60 | 69.82                 | 18.73 | 13.26 |

Table 12: Dropout reconstruction with Guided Backprop on the CiteSeer dataset using the GCN model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |      | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7  | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 21.74                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 2.41                  | 144.27                | 216.10                | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.26                  | 115.87                | 142.42                | 4.39                         | 0.57  | 0.24 | 0.47                  | 0.04 | 0.02 |
|                       | 0.50            | 1.94                  | 72.35                 | 51.34                 | 8.21                         | 1.14  | 0.33 | 1.03                  | 0.08 | 0.02 |
|                       | 0.70            | 1.81                  | 42.41                 | 16.83                 | 9.53                         | 1.68  | 0.57 | 1.38                  | 0.12 | 0.04 |
|                       | 1.00            | 1.69                  | 0.00                  | -6.72                 | 11.72                        | 2.31  | 0.81 | 1.68                  | 0.17 | 0.05 |
| 32                    | 0.00            | 1.58                  | 244.27                | 372.58                | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.44                  | 195.16                | 240.14                | 4.78                         | 0.60  | 0.12 | 0.52                  | 0.04 | 0.01 |
|                       | 0.50            | 1.22                  | 123.47                | 95.47                 | 10.37                        | 1.32  | 0.18 | 1.40                  | 0.09 | 0.01 |
|                       | 0.70            | 1.15                  | 71.30                 | 27.37                 | 13.20                        | 1.95  | 0.48 | 1.88                  | 0.14 | 0.03 |
|                       | 1.00            | 0.96                  | 0.00                  | -9.47                 | 15.18                        | 2.58  | 0.57 | 2.27                  | 0.18 | 0.04 |
| 64                    | 0.00            | 1.18                  | 383.68                | 582.29                | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.16                  | 305.98                | 374.43                | 7.42                         | 0.63  | 0.09 | 0.78                  | 0.04 | 0.01 |
|                       | 0.50            | 0.95                  | 191.88                | 144.68                | 12.44                        | 1.80  | 0.51 | 1.82                  | 0.14 | 0.03 |
|                       | 0.70            | 0.86                  | 116.65                | 48.55                 | 15.96                        | 2.71  | 0.78 | 2.57                  | 0.20 | 0.05 |
|                       | 1.00            | 0.77                  | 0.00                  | -13.16                | 18.52                        | 3.40  | 1.05 | 3.23                  | 0.25 | 0.07 |
| 128                   | 0.00            | 1.24                  | 647.40                | 1012.32               | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.28                  | 518.52                | 649.45                | 14.97                        | 3.58  | 1.20 | 1.93                  | 0.26 | 0.08 |
|                       | 0.50            | 1.12                  | 325.04                | 247.10                | 24.56                        | 8.00  | 2.92 | 4.63                  | 0.61 | 0.20 |
|                       | 0.70            | 0.97                  | 192.82                | 72.69                 | 28.13                        | 10.43 | 3.31 | 6.40                  | 0.85 | 0.23 |
|                       | 1.00            | 0.85                  | 0.00                  | -29.09                | 32.37                        | 14.43 | 5.50 | 8.41                  | 1.34 | 0.40 |

Table 13: Dropout reconstruction with Saliency Map on the CiteSeer dataset using the GCN model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |      | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7  | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 21.96                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 2.39                  | 144.27                | 216.10                | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.18                  | 115.57                | 137.85                | 3.61                         | 0.36  | 0.12 | 0.37                  | 0.03 | 0.01 |
|                       | 0.50            | 1.96                  | 70.33                 | 48.57                 | 8.51                         | 1.23  | 0.36 | 1.07                  | 0.08 | 0.02 |
|                       | 0.70            | 1.80                  | 41.33                 | 14.96                 | 10.31                        | 1.62  | 0.51 | 1.41                  | 0.11 | 0.03 |
|                       | 1.00            | 1.64                  | 0.00                  | -6.72                 | 11.75                        | 2.31  | 0.81 | 1.68                  | 0.17 | 0.05 |
| 32                    | 0.00            | 1.56                  | 244.27                | 372.58                | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.47                  | 195.40                | 244.90                | 4.90                         | 0.51  | 0.06 | 0.55                  | 0.04 | 0.00 |
|                       | 0.50            | 1.19                  | 124.17                | 92.60                 | 9.95                         | 1.41  | 0.33 | 1.39                  | 0.10 | 0.02 |
|                       | 0.70            | 1.11                  | 72.47                 | 28.10                 | 12.92                        | 1.98  | 0.39 | 1.87                  | 0.14 | 0.03 |
|                       | 1.00            | 0.95                  | 0.00                  | -9.47                 | 15.00                        | 2.55  | 0.57 | 2.26                  | 0.18 | 0.04 |
| 64                    | 0.00            | 1.17                  | 383.68                | 582.29                | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.15                  | 306.40                | 366.48                | 6.55                         | 0.75  | 0.09 | 0.73                  | 0.05 | 0.01 |
|                       | 0.50            | 0.96                  | 193.00                | 141.83                | 13.89                        | 1.86  | 0.45 | 1.96                  | 0.14 | 0.03 |
|                       | 0.70            | 0.83                  | 112.32                | 43.63                 | 16.20                        | 2.68  | 0.66 | 2.62                  | 0.19 | 0.04 |
|                       | 1.00            | 0.77                  | 0.00                  | -13.16                | 18.27                        | 3.49  | 0.93 | 3.22                  | 0.26 | 0.06 |
| 128                   | 0.00            | 1.28                  | 647.40                | 1012.32               | 0.00                         | 0.00  | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.26                  | 513.53                | 628.51                | 13.29                        | 3.28  | 1.23 | 1.81                  | 0.23 | 0.08 |
|                       | 0.50            | 1.12                  | 326.21                | 250.59                | 24.05                        | 7.81  | 2.46 | 4.49                  | 0.60 | 0.17 |
|                       | 0.70            | 0.97                  | 194.17                | 76.71                 | 27.83                        | 10.31 | 3.46 | 6.13                  | 0.82 | 0.24 |
|                       | 1.00            | 0.84                  | 0.00                  | -29.09                | 32.25                        | 14.58 | 5.65 | 8.29                  | 1.36 | 0.41 |

Table 14: Dropout reconstruction with DeconvNet on the CiteSeer dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 35.61                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 3.84                  | 144.27                | 216.10                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 3.55                  | 115.09                | 132.07                | 4.78                         | 3.49  | 2.98  | 1.14                  | 0.56 | 0.41 |
|                       | 0.50            | 3.24                  | 70.24                 | 50.92                 | 10.52                        | 7.60  | 6.70  | 2.32                  | 1.16 | 0.86 |
|                       | 0.70            | 3.05                  | 44.88                 | 16.72                 | 13.07                        | 9.41  | 8.66  | 3.09                  | 1.53 | 1.14 |
|                       | 1.00            | 2.93                  | 0.00                  | -6.72                 | 15.06                        | 10.79 | 9.59  | 3.46                  | 1.72 | 1.27 |
| 32                    | 0.00            | 2.34                  | 244.27                | 372.58                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.04                  | 194.77                | 232.29                | 4.57                         | 2.86  | 2.01  | 0.92                  | 0.38 | 0.25 |
|                       | 0.50            | 1.80                  | 121.07                | 88.14                 | 13.41                        | 9.65  | 8.36  | 3.00                  | 1.46 | 1.07 |
|                       | 0.70            | 1.62                  | 71.63                 | 28.89                 | 15.03                        | 11.18 | 9.68  | 3.54                  | 1.77 | 1.25 |
|                       | 1.00            | 1.43                  | 0.00                  | -9.47                 | 19.36                        | 14.73 | 12.71 | 4.60                  | 2.37 | 1.73 |
| 64                    | 0.00            | 1.63                  | 383.68                | 582.29                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.51                  | 306.76                | 368.91                | 6.07                         | 3.85  | 3.19  | 1.25                  | 0.57 | 0.40 |
|                       | 0.50            | 1.24                  | 194.35                | 145.65                | 13.83                        | 9.32  | 8.30  | 2.94                  | 1.43 | 1.05 |
|                       | 0.70            | 1.17                  | 115.72                | 48.95                 | 18.43                        | 13.44 | 11.69 | 4.24                  | 2.12 | 1.57 |
|                       | 1.00            | 0.98                  | 0.00                  | -13.16                | 23.47                        | 17.67 | 15.63 | 5.46                  | 2.81 | 2.12 |
| 128                   | 0.00            | 1.50                  | 647.40                | 1012.32               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.54                  | 520.17                | 663.22                | 11.39                        | 7.06  | 5.95  | 2.28                  | 0.98 | 0.70 |
|                       | 0.50            | 1.25                  | 323.53                | 247.80                | 22.45                        | 16.11 | 13.89 | 4.88                  | 2.38 | 1.71 |
|                       | 0.70            | 1.12                  | 195.58                | 85.52                 | 26.99                        | 20.08 | 18.18 | 6.25                  | 3.26 | 2.41 |
|                       | 1.00            | 0.99                  | 0.00                  | -29.09                | 33.36                        | 26.54 | 24.02 | 7.99                  | 4.30 | 3.23 |

Table 15: Dropout reconstruction with GNNExplainer on the CiteSeer dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |       |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|-------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5   | 0.7  |
| 1                     | 0.00            | 3816.54               | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00  | 0.00 |
| 16                    | 0.00            | 325.03                | 144.27                | 216.10                | 66.55                        | 26.57 | 0.00  | 13.86                 | 3.45  | 0.00 |
|                       | 0.20            | 302.25                | 116.56                | 142.16                | 70.33                        | 30.81 | 7.78  | 23.21                 | 4.68  | 0.82 |
|                       | 0.50            | 271.80                | 72.02                 | 50.33                 | 71.84                        | 32.55 | 12.05 | 28.15                 | 6.08  | 1.80 |
|                       | 0.70            | 254.67                | 43.31                 | 15.42                 | 71.84                        | 33.78 | 15.90 | 34.05                 | 6.75  | 2.51 |
|                       | 1.00            | 241.96                | 0.00                  | -6.72                 | 72.71                        | 34.45 | 17.22 | 38.26                 | 7.59  | 3.06 |
| 32                    | 0.00            | 197.18                | 244.27                | 372.58                | 67.21                        | 27.44 | 0.06  | 13.88                 | 3.39  | 0.00 |
|                       | 0.20            | 174.57                | 196.57                | 240.95                | 71.84                        | 31.71 | 7.69  | 21.70                 | 4.84  | 0.98 |
|                       | 0.50            | 149.68                | 123.87                | 94.00                 | 73.52                        | 33.15 | 13.71 | 33.79                 | 6.27  | 2.23 |
|                       | 0.70            | 135.41                | 74.60                 | 29.70                 | 72.32                        | 34.51 | 17.94 | 40.75                 | 7.50  | 3.25 |
|                       | 1.00            | 121.05                | 0.00                  | -9.47                 | 72.62                        | 35.86 | 21.19 | 46.81                 | 8.63  | 4.09 |
| 64                    | 0.00            | 124.11                | 383.68                | 582.29                | 66.91                        | 27.95 | 0.00  | 13.85                 | 3.54  | 0.00 |
|                       | 0.20            | 104.76                | 307.15                | 378.41                | 72.23                        | 31.41 | 8.93  | 25.59                 | 5.34  | 1.30 |
|                       | 0.50            | 82.37                 | 191.70                | 138.62                | 73.52                        | 34.30 | 17.13 | 40.84                 | 6.94  | 2.94 |
|                       | 0.70            | 72.12                 | 117.73                | 49.47                 | 72.86                        | 34.33 | 19.09 | 46.53                 | 8.63  | 4.11 |
|                       | 1.00            | 59.48                 | 0.00                  | -13.16                | 70.78                        | 36.13 | 23.32 | 54.39                 | 9.96  | 5.38 |
| 128                   | 0.00            | 85.74                 | 647.40                | 1012.32               | 67.51                        | 27.68 | 0.00  | 13.84                 | 3.51  | 0.00 |
|                       | 0.20            | 71.41                 | 518.30                | 647.98                | 74.51                        | 35.83 | 17.10 | 30.99                 | 6.50  | 2.29 |
|                       | 0.50            | 51.81                 | 324.11                | 242.40                | 73.22                        | 38.35 | 25.22 | 50.29                 | 9.68  | 5.20 |
|                       | 0.70            | 41.19                 | 197.51                | 79.06                 | 71.99                        | 39.86 | 28.67 | 59.99                 | 11.89 | 7.08 |
|                       | 1.00            | 30.59                 | 0.00                  | -29.09                | 70.87                        | 42.26 | 33.18 | 69.22                 | 14.09 | 9.12 |

Table 16: Dropout reconstruction with Guided Backprop on the CiteSeer dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 36.03                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 3.53                  | 144.27                | 216.10                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 3.30                  | 114.07                | 135.90                | 3.40                         | 2.22  | 2.04  | 0.80                  | 0.39 | 0.28 |
|                       | 0.50            | 3.03                  | 72.77                 | 51.41                 | 11.51                        | 8.54  | 7.54  | 2.65                  | 1.32 | 0.96 |
|                       | 0.70            | 2.97                  | 43.37                 | 17.60                 | 9.53                         | 5.98  | 5.02  | 1.92                  | 0.87 | 0.63 |
|                       | 1.00            | 2.86                  | 0.00                  | -6.72                 | 15.06                        | 10.76 | 9.56  | 3.46                  | 1.72 | 1.27 |
| 32                    | 0.00            | 2.39                  | 244.27                | 372.58                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.20                  | 196.60                | 248.51                | 3.94                         | 2.49  | 1.95  | 0.68                  | 0.33 | 0.24 |
|                       | 0.50            | 1.78                  | 122.15                | 90.75                 | 13.80                        | 10.82 | 9.92  | 3.41                  | 1.72 | 1.29 |
|                       | 0.70            | 1.66                  | 74.84                 | 32.56                 | 12.14                        | 8.72  | 7.21  | 2.84                  | 1.37 | 0.95 |
|                       | 1.00            | 1.45                  | 0.00                  | -9.47                 | 19.33                        | 14.70 | 12.71 | 4.59                  | 2.37 | 1.72 |
| 64                    | 0.00            | 1.59                  | 383.68                | 582.29                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.49                  | 307.91                | 380.89                | 7.42                         | 4.81  | 4.18  | 1.64                  | 0.76 | 0.58 |
|                       | 0.50            | 1.29                  | 192.34                | 149.34                | 14.52                        | 10.07 | 8.72  | 3.14                  | 1.48 | 1.13 |
|                       | 0.70            | 1.13                  | 116.35                | 50.53                 | 19.63                        | 14.25 | 12.89 | 4.71                  | 2.35 | 1.74 |
|                       | 1.00            | 1.00                  | 0.00                  | -13.16                | 23.44                        | 17.64 | 15.63 | 5.46                  | 2.81 | 2.13 |
| 128                   | 0.00            | 1.51                  | 647.40                | 1012.32               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.52                  | 519.09                | 663.03                | 8.75                         | 4.51  | 3.88  | 1.58                  | 0.64 | 0.46 |
|                       | 0.50            | 1.27                  | 323.87                | 247.43                | 21.52                        | 15.12 | 13.32 | 4.62                  | 2.29 | 1.70 |
|                       | 0.70            | 1.16                  | 192.73                | 76.49                 | 27.41                        | 20.47 | 17.70 | 6.15                  | 3.10 | 2.30 |
|                       | 1.00            | 0.97                  | 0.00                  | -29.09                | 33.27                        | 26.51 | 23.96 | 7.98                  | 4.30 | 3.22 |

Table 17: Dropout reconstruction with Saliency Map on the CiteSeer dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 35.65                 | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 3.57                  | 144.27                | 216.10                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 3.28                  | 116.95                | 141.10                | 2.74                         | 1.41  | 1.08  | 0.49                  | 0.17 | 0.11 |
|                       | 0.50            | 2.95                  | 71.51                 | 48.68                 | 9.86                         | 6.55  | 5.80  | 2.17                  | 1.00 | 0.73 |
|                       | 0.70            | 2.77                  | 44.63                 | 18.26                 | 13.23                        | 9.35  | 8.24  | 2.93                  | 1.43 | 1.03 |
|                       | 1.00            | 2.62                  | 0.00                  | -6.72                 | 15.21                        | 10.88 | 9.65  | 3.45                  | 1.69 | 1.26 |
| 32                    | 0.00            | 2.24                  | 244.27                | 372.58                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 2.04                  | 194.89                | 227.68                | 6.31                         | 4.90  | 4.42  | 1.30                  | 0.67 | 0.51 |
|                       | 0.50            | 1.81                  | 123.93                | 93.48                 | 10.55                        | 7.42  | 6.46  | 2.43                  | 1.13 | 0.80 |
|                       | 0.70            | 1.62                  | 73.37                 | 29.31                 | 14.91                        | 10.70 | 9.44  | 3.33                  | 1.63 | 1.17 |
|                       | 1.00            | 1.43                  | 0.00                  | -9.47                 | 19.36                        | 14.82 | 13.07 | 4.52                  | 2.31 | 1.67 |
| 64                    | 0.00            | 1.62                  | 383.68                | 582.29                | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.48                  | 307.94                | 366.98                | 7.12                         | 5.05  | 4.57  | 1.65                  | 0.76 | 0.56 |
|                       | 0.50            | 1.23                  | 191.16                | 142.07                | 14.64                        | 10.04 | 8.81  | 3.23                  | 1.51 | 1.08 |
|                       | 0.70            | 1.15                  | 116.71                | 50.70                 | 18.12                        | 12.89 | 11.57 | 4.13                  | 2.01 | 1.47 |
|                       | 1.00            | 0.96                  | 0.00                  | -13.16                | 23.69                        | 17.73 | 15.87 | 5.45                  | 2.76 | 2.07 |
| 128                   | 0.00            | 1.56                  | 647.40                | 1012.32               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 1.53                  | 517.64                | 649.36                | 11.21                        | 7.18  | 6.04  | 2.29                  | 1.00 | 0.74 |
|                       | 0.50            | 1.25                  | 322.03                | 243.06                | 22.90                        | 16.53 | 14.34 | 5.03                  | 2.40 | 1.77 |
|                       | 0.70            | 1.13                  | 193.45                | 77.33                 | 27.65                        | 20.95 | 18.88 | 6.48                  | 3.24 | 2.37 |
|                       | 1.00            | 0.95                  | 0.00                  | -29.09                | 33.36                        | 26.69 | 24.17 | 7.94                  | 4.21 | 3.16 |

Table 18: Dropout reconstruction with DeconvNet on the PubMed dataset using the GCN model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |      |      | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|------|------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5  | 0.7  | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 283.01                | 0.00                  | 0.00                  | 0.00                         | 0.00 | 0.00 | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 133.09                | 875.45                | 1020.18               | 0.00                         | 0.00 | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 96.07                 | 701.05                | 654.14                | 9.91                         | 0.47 | 0.07 | 0.35                  | 0.01 | 0.00 |
|                       | 0.50            | 53.63                 | 436.46                | 252.70                | 17.68                        | 1.20 | 0.19 | 0.81                  | 0.02 | 0.00 |
|                       | 0.70            | 34.68                 | 261.85                | 82.38                 | 21.03                        | 1.87 | 0.23 | 1.11                  | 0.04 | 0.00 |
|                       | 1.00            | 17.80                 | 0.00                  | -16.70                | 24.19                        | 2.27 | 0.34 | 1.26                  | 0.04 | 0.01 |

Table 19: Dropout reconstruction with Guided Backprop on the PubMed dataset using the GCN model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |      |      | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|------|------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5  | 0.7  | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 283.40                | 0.00                  | 0.00                  | 0.00                         | 0.00 | 0.00 | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 133.50                | 875.45                | 1020.18               | 0.00                         | 0.00 | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 96.65                 | 700.08                | 651.77                | 9.63                         | 0.55 | 0.08 | 0.34                  | 0.01 | 0.00 |
|                       | 0.50            | 54.00                 | 437.61                | 251.60                | 17.19                        | 1.35 | 0.16 | 0.79                  | 0.03 | 0.00 |
|                       | 0.70            | 34.75                 | 261.46                | 84.25                 | 21.56                        | 1.83 | 0.25 | 1.12                  | 0.03 | 0.00 |
|                       | 1.00            | 18.26                 | 0.00                  | -16.70                | 24.20                        | 2.28 | 0.34 | 1.26                  | 0.04 | 0.01 |

Table 20: Dropout reconstruction with Saliency Map on the PubMed dataset using the GCN model

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |      |      | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|------|------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5  | 0.7  | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 294.67                | 0.00                  | 0.00                  | 0.00                         | 0.00 | 0.00 | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 132.81                | 875.45                | 1020.18               | 0.00                         | 0.00 | 0.00 | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 95.81                 | 699.60                | 650.67                | 9.60                         | 0.50 | 0.07 | 0.32                  | 0.01 | 0.00 |
|                       | 0.50            | 53.65                 | 438.81                | 248.93                | 16.91                        | 1.32 | 0.20 | 0.76                  | 0.02 | 0.00 |
|                       | 0.70            | 34.38                 | 261.90                | 83.51                 | 21.51                        | 1.97 | 0.30 | 1.09                  | 0.04 | 0.01 |
|                       | 1.00            | 17.79                 | 0.00                  | -16.70                | 24.05                        | 2.42 | 0.39 | 1.22                  | 0.05 | 0.01 |

Table 21: Dropout reconstruction with DeconvNet on the PubMed dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 403.32                | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 182.39                | 875.45                | 1020.18               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 132.23                | 700.79                | 648.32                | 16.07                        | 6.44  | 4.13  | 0.68                  | 0.18 | 0.09 |
|                       | 0.50            | 75.30                 | 438.18                | 251.91                | 29.48                        | 13.71 | 9.59  | 1.65                  | 0.43 | 0.23 |
|                       | 0.70            | 48.17                 | 262.44                | 82.84                 | 35.72                        | 18.18 | 13.00 | 2.22                  | 0.60 | 0.33 |
|                       | 1.00            | 25.03                 | 0.00                  | -16.70                | 41.61                        | 21.70 | 16.18 | 2.78                  | 0.76 | 0.42 |

Table 22: Dropout reconstruction with Guided Backprop on the PubMed dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 399.74                | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 182.89                | 875.45                | 1020.18               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 132.46                | 700.33                | 651.08                | 16.07                        | 6.33  | 4.12  | 0.72                  | 0.18 | 0.10 |
|                       | 0.50            | 75.51                 | 437.71                | 252.07                | 29.30                        | 13.67 | 9.71  | 1.68                  | 0.44 | 0.24 |
|                       | 0.70            | 48.27                 | 264.23                | 85.06                 | 34.61                        | 17.41 | 12.47 | 2.13                  | 0.58 | 0.31 |
|                       | 1.00            | 25.64                 | 0.00                  | -16.70                | 41.61                        | 21.70 | 16.18 | 2.78                  | 0.76 | 0.42 |

Table 23: Dropout reconstruction with Saliency Map on the PubMed dataset using the GAT model.

| Number<br>of clusters | Dropout<br>rate | Execution<br>time (s) | $\Delta$ nodes<br>(%) | $\Delta$ edges<br>(%) | Affected<br>explanations (%) |       |       | Affected<br>nodes (%) |      |      |
|-----------------------|-----------------|-----------------------|-----------------------|-----------------------|------------------------------|-------|-------|-----------------------|------|------|
|                       |                 |                       |                       |                       | 0.2                          | 0.5   | 0.7   | 0.2                   | 0.5  | 0.7  |
| 1                     | 0.00            | 401.21                | 0.00                  | 0.00                  | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
| 16                    | 0.00            | 182.07                | 875.45                | 1020.18               | 0.00                         | 0.00  | 0.00  | 0.00                  | 0.00 | 0.00 |
|                       | 0.20            | 132.18                | 701.16                | 653.18                | 15.97                        | 6.31  | 4.22  | 0.71                  | 0.19 | 0.11 |
|                       | 0.50            | 75.33                 | 440.30                | 250.29                | 30.67                        | 14.18 | 9.93  | 1.71                  | 0.45 | 0.24 |
|                       | 0.70            | 48.14                 | 263.62                | 83.85                 | 35.62                        | 17.55 | 12.81 | 2.19                  | 0.59 | 0.33 |
|                       | 1.00            | 25.10                 | 0.00                  | -16.70                | 41.65                        | 21.61 | 16.13 | 2.78                  | 0.76 | 0.42 |